Department of Electrical and Electronics Engineering
Academic Year 2022 – 2023 (Even Semester)

Degree, Semester & Branch: VIII Semester B.E EEE
Course Code & Title: EE8019 – Smart Grid
Name of the Faculty member (s): Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic: Unit I / National Initiatives in Smart Grid**
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 4
- **Activity Chosen: Case Study Report**
- **Justification:**
  - The chosen topic will be of facts and figures about the pilot projects in India. The current status of the Smart Grid projects needs to be checked and presented. If all these data are presented in the conventional class room lecture, the students may not be interested and may not understand fully.
  - If the students are encouraged to present these data in any Info graphic form / table form, then they will understand the concepts better and also enjoy the session. That is the reason for choosing this activity "Case Study Report".
- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**
  - The students are taken to the Power System Simulation Laboratory and they are provided with individual computer system with internet connectivity.
  - The components/technologies of the smart grid and the Smart Grid pilot projects of the India have been briefly explained to the students.
  - The students are asked to prepare the case study report on the topic Smart Grid Initiatives in India.
  - The guidelines for the contents of the report and formatting of the report have been told to the students.
  - At the end of the period, the students have submitted their report in the CANVAS – LMS.
- **CO – PO / PSO mapping:**

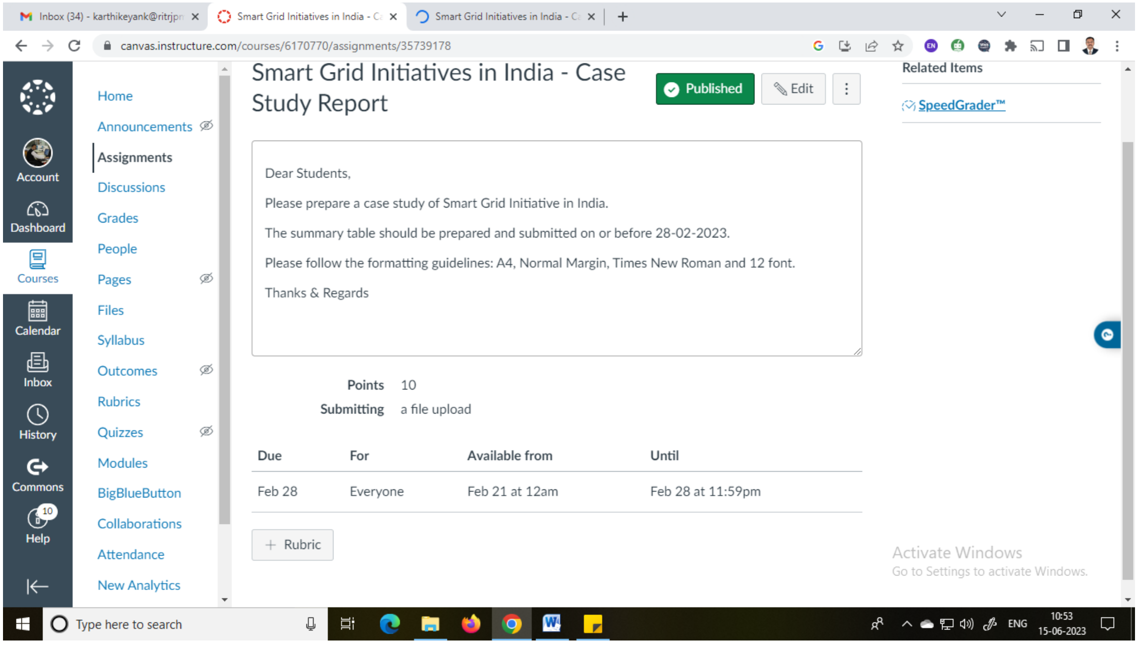
CO	PO 1	PO 2	PO 5	PO 12	PSO 1
CO2	3	1	1	1	1

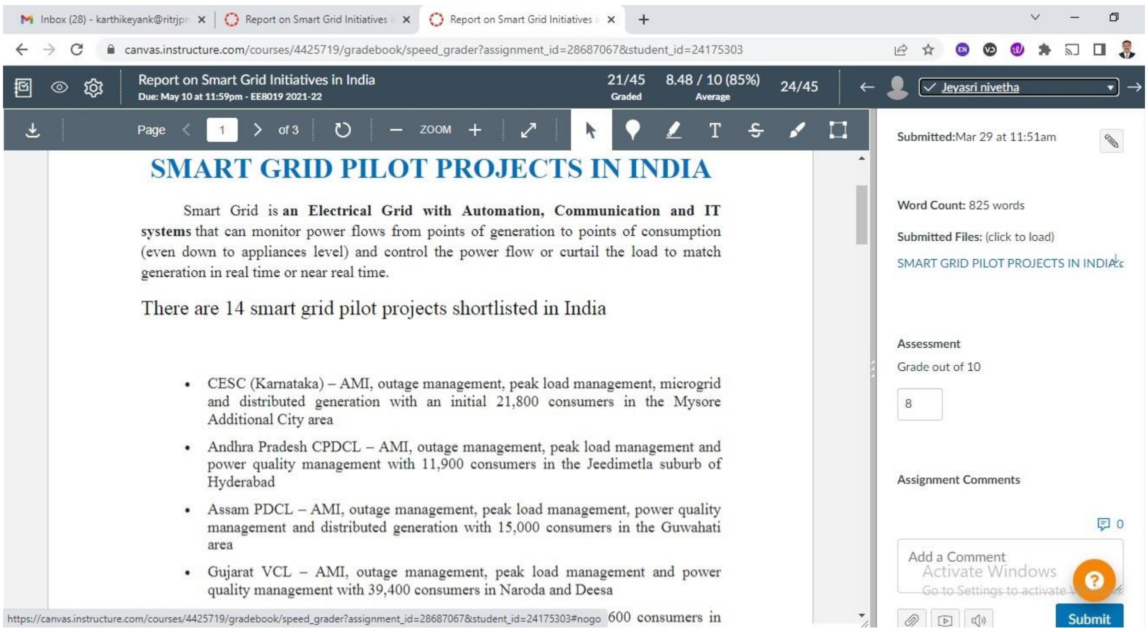
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 5**	**PO 8**	**PO 10**
	1	**1**	**1**
Justification for correlation	The students will be using the internet, drawing tools etc., for preparing the report.	The students will follow the ethics and they will not be allowed to copy from other students.	The students are writing the report; hence their written communication is improved.

- **Images / Screenshot of the practice:**





- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - ✓ Many students requested to conduct this type of activity frequently, because it provided them an opportunity to have hands on training with MS-word and some drawing tool.
  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ The students have prepared their report and hence they got better understanding about the concept, it was evident once I asked few to comprehend the points written in their report.
    - ✓ The student's presentation skills got improved since they prepared report and it was also useful for their exam preparation.
  - ***Challenges faced in implementation:***
    - ✓ Initially the students hesitated to prepare a report. I explained them to understand the work they need to complete and helped them in doing so in all the phases.
    - ✓ Managing the non performing students was quite difficult. They used to login some other websites and not doing the task.

References:

- I.S. Jha, Subir Sen, Rajesh Kumar, and D.P. Kothari, "Smart Grid Fundamentals & Applications", New Age International Publishers, 1st Edition, 2019.
- Bharat Modi, Anu Prakash, and Yogesh Kumar, "Fundamentals of Smart Grid Technology", S. K. Kataria & Sons, 1st Edition, 2019.
- https://www.iitk.ac.in/npsc/Papers/NPSC2014/1569993451.pdf
- https://indiasmartgrid.org/nsgm_about.php
- https://events.development.asia/system/files/materials/2018/04/201804-smart-grid-developments-india.pdf

Signature of Faculty Member **HoD/EEE**

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Innovative Practice Description

- **Unit / Topic: Unit II / Plugin Hybrid Electric Vehicles (PHEV)**
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 9
- **Activity Chosen:** Think – Pair - Share
- **Justification:**
  - The concept of Electric Vehicle is relatively a new concept for the students. Especially the operation of IC engine based vehicles completely new to the EEE students. The PHEV is a hybrid of IC engine and EV and hence the students may find it difficult to understand the concepts. Therefore, the team activity "Think-Pair-Share" has been planned for this topic.
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
  - The students have been grouped into smaller groups of four members, with one bright student, two average learners and one slow learner in each group. One average learner from each group has been made as a recorder.
  - I have shown the power point presentation for the quick recap of the concepts like Vehicle types, benefits of Electric Vehicle, and the constructional difference between IC engine vehicle and Electric Vehicle.
  - The detailed operation procedure of the IC engine vehicle has been explained to the students.
  - The students have been told to recollect the concepts of Plug-in Hybrid Vehicle (PHEV) and then asked to discuss about it within the group for 5 minutes.
  - Finally, two recorders have been asked to share their results to the whole class through chalk and talk technique. All other students interacted with them for proper understanding the concepts.

- **CO – PO / PSO mapping:**

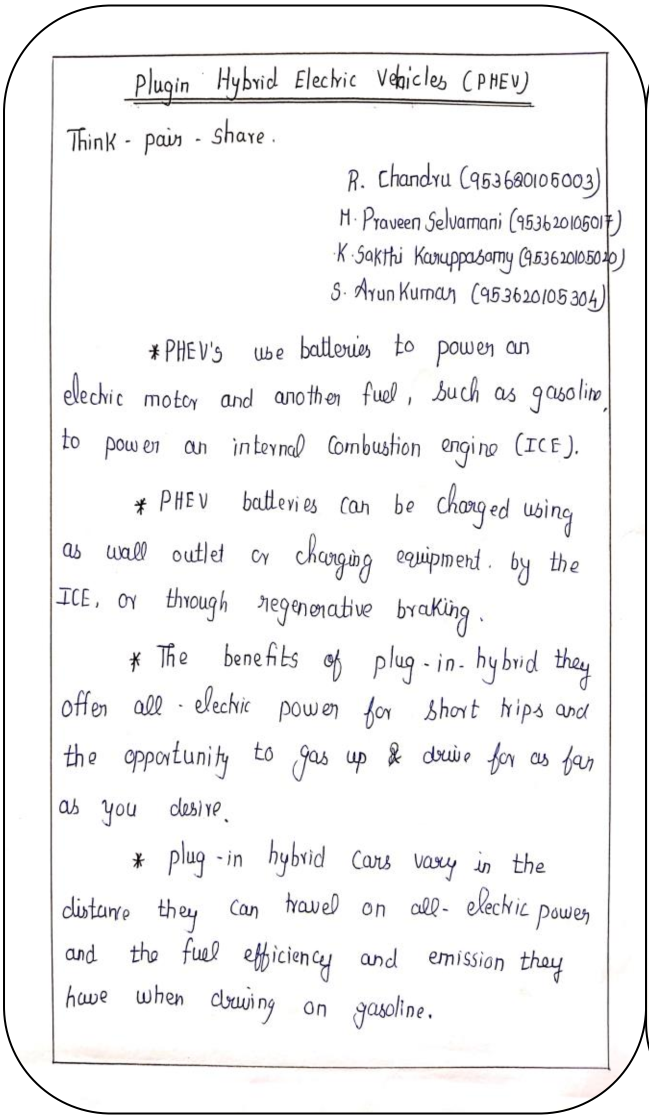
CO	PO 1	PO 2	PO 5	PO 8	PO12	PSO 1
CO2	3	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 9**	**PO 10**
	1	**1**
Justification for correlation	The students will do this activity in team. Hence it is mapped with PO 9 with level 1.	The students will communicate with each other and do the oral presentation. Hence this activity is mapped with PO 10 with level 1.

- **Images / Screenshot of the practice:**



Plugin Hybrid Electric Vehicles (PHEV)

Think - pair - share.

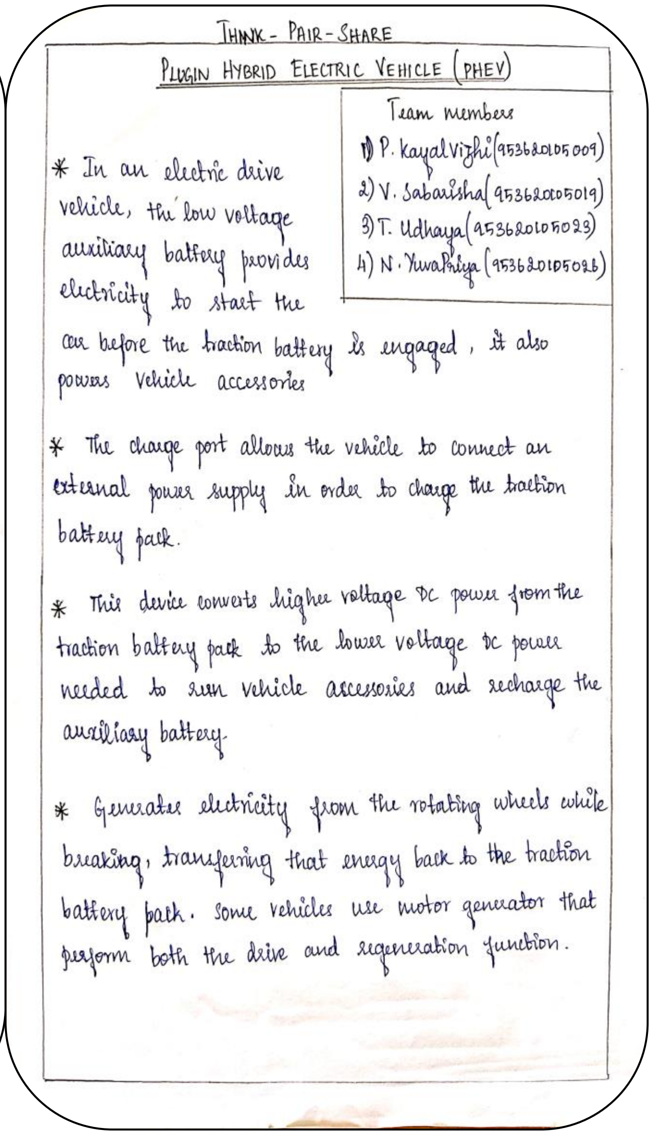
R. Chandru (953680105003)
M. Praveen Selvamani (953620105017)
K. Sakthi Karuppasamy (953620105020)
S. Arun Kumar (953620105304)

* PHEV's use batteries to power an electric motor and another fuel, such as gasoline, to power an internal combustion engine (ICE).

* PHEV batteries can be charged using as wall outlet or charging equipment. by the ICE, or through regenerative braking.

* The benefits of plug-in hybrid they offer all-electric power for short trips and the opportunity to gas up & drive for as far as you desire.

* plug-in hybrid cars vary in the distance they can travel on all-electric power and the fuel efficiency and emission they have when driving on gasoline.



THINK - PAIR - SHARE

PLUGIN HYBRID ELECTRIC VEHICLE (PHEV)

Team members
1) P. Kayalvizhi (953620105009)
2) V. Sabarisha (953620105019)
3) T. Udhaya (953620105023)
4) N. YuvaPriya (953620105026)

* In an electric drive vehicle, the low voltage auxiliary battery provides electricity to start the car before the traction battery is engaged, it also powers vehicle accessories

* The charge port allows the vehicle to connect an external power supply in order to charge the traction battery pack.

* This device converts higher voltage DC power from the traction battery pack to the lower voltage DC power needed to run vehicle accessories and recharge the auxiliary battery.

* Generates electricity from the rotating wheels while breaking, transferring that energy back to the traction battery pack. Some vehicles use motor generator that perform both the drive and regeneration function.

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - ✓ The students have enjoyed the activity. Most of the students requested me to conduct such activities in the classes. When I did this activity, they had a great confidence in learning the new concept.
  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ The students have understood the concepts clearly and it has been evident from the points they have shared with the classroom.
    - ✓ The students have presented the points to the class; it cleared any ambiguity in understanding the important points in the chosen topic.
  - ***Challenges faced in implementation:***
    - ✓ The activity has been planned for 15 minutes, but it has been completed after 16 minutes only. Next time, I should plan the activity with proper time.
    - ✓ The group has to form as heterogeneous, and then only I can take care about the non-participating students.

References:

- ❖ I.S. Jha, Subir Sen, Rajesh Kumar, and D.P. Kothari, "Smart Grid Fundamentals & Applications", New Age International Publishers, 1st Edition, 2019.
- ❖ Bharat Modi, Anu Prakash, and Yogesh Kumar, "Fundamentals of Smart Grid Technology", S. K. Kataria & Sons, 1st Edition, 2019.
- ❖ https://afdc.energy.gov/vehicles/electric_basics_phev.html

Signature of Faculty Member **HoD/EEE**

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

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Innovative Practice Description

- **Unit / Topic: Unit III / Advanced Metering infrastructure (AMI) drivers and benefits**
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 11
- **Activity Chosen: 3-2-1 Method**
- **Justification:**
  - This strategy provides a structure for students to record their own comprehension and summarize their learning. It also gives teachers the opportunity to identify areas that need re-teaching, as well as areas of student interest and or background.
  - It can be used as a closing activity so that students can review what was learned in the lesson.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
  - The students are told to write 3-points about needs of the AMI, 2-points about benefits of the AMI and 1-point about the standards of the AMI in smart grid in a paper.
  - They are allowed to discuss about the points they have written with other students and instructor and fine tune their points.
  - Few students have been called to explain they have written on their paper.
  - Other students are instructed to listen and raise their queries to the instructor.
- **CO – PO / PSO mapping:**

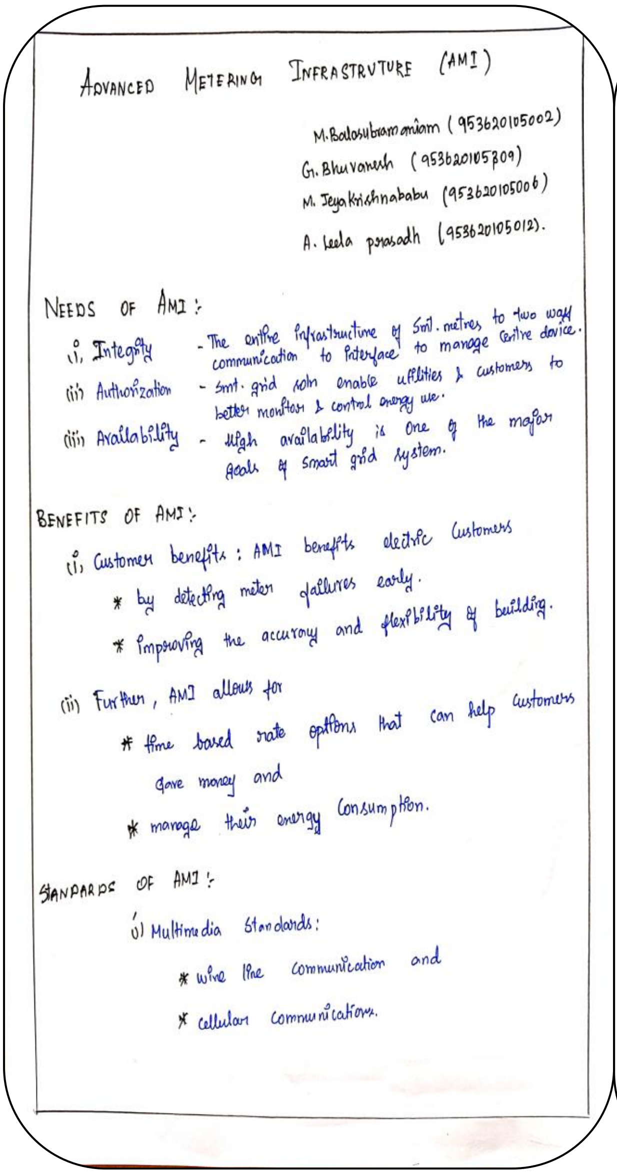
CO	PO 1	PO 2	PO 5	PO 12	PSO 1
CO2	3	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 8	PO 10
	1	1
Justification for correlation	Students will follow the basic ethical practices while doing this activity. The students should not copy the points from the neighboring students.	Students will explain about the points noted by them while presenting their 3-2-1 points to the class. It improves their oral communication.

- **Images / Screenshot of the practice:**


ADVANCED METERING INFRASTRUCTURE (AMI)

M.Balasubramaniam (953620105002)
G. Bhuvanesh (953620105309)
M. Jeyakrishnababu (953620105006)
A. Leela prasadh (953620105012).

NEEDS OF AMI :-

(i) Integrity - The entire infrastructure of Smt. metres to two way communication to interface to manage centre device.
(ii) Authorization - Smt. grid soln enable utilities & customers to better monitor & control energy use.
(iii) Availability - High availability is one of the major goals of smart grid system.

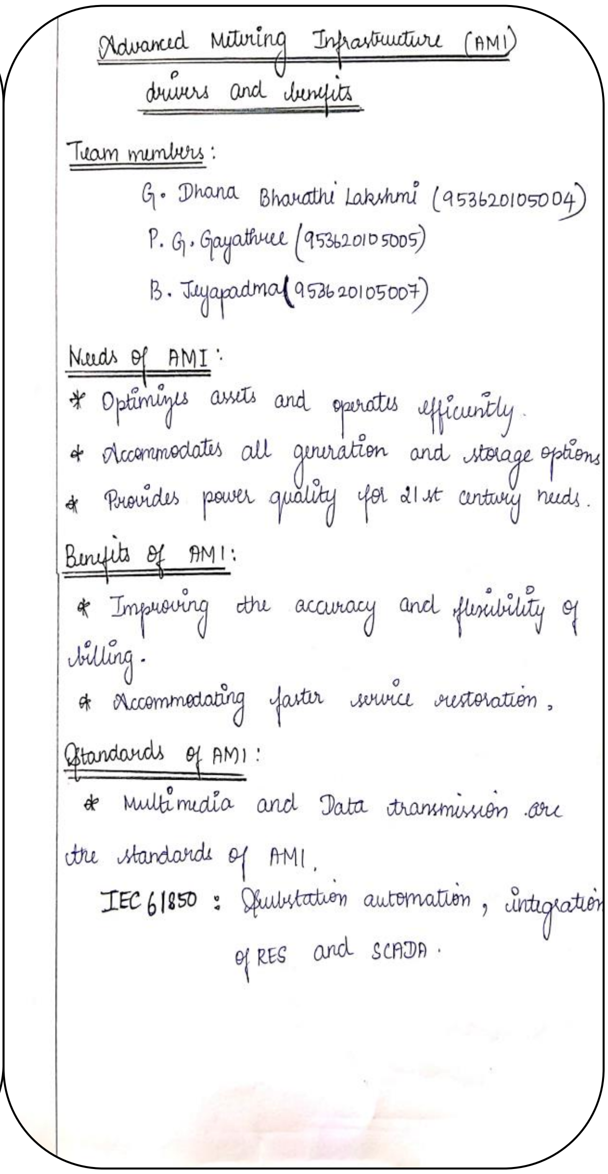
BENEFITS OF AMI :-

(i) Customer benefits : AMI benefits electric customers
* by detecting meter failures early.
* Improving the accuracy and flexibility of building.

(ii) Further, AMI allows for
* time based rate options that can help customers save money and
* manage their energy consumption.

STANDARDS OF AMI :-

(i) Multimedia Standards:
* wire line communication and
* cellular communications.

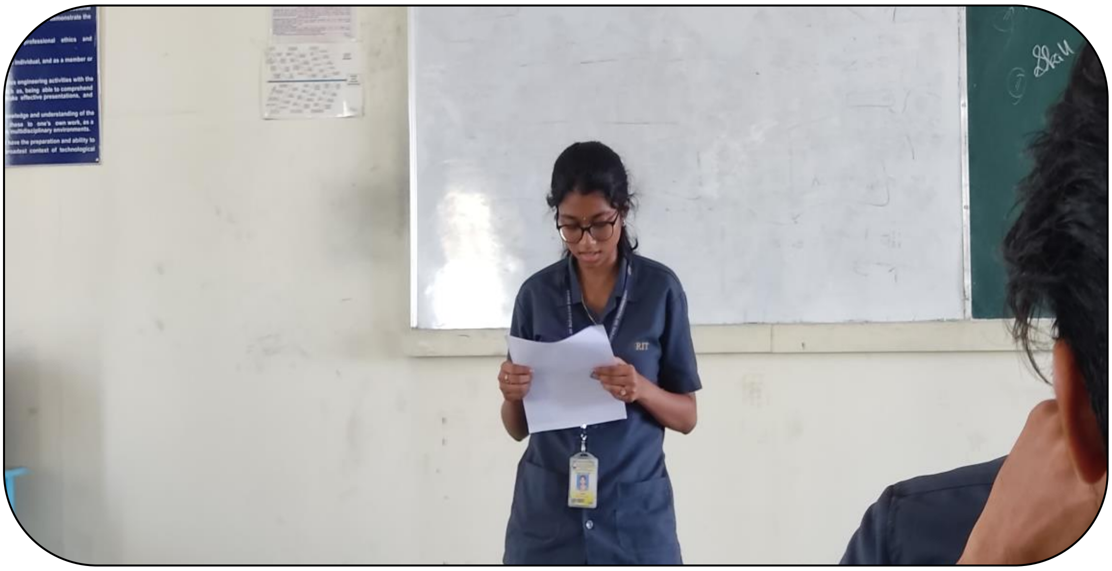

Advanced Metering Infrastructure (AMI) drivers and benefits

Team members :
G. Dhana Bharathi Lakshmi (953620105004)
P. G. Gayathree (953620105005)
B. Jeyapadma (953620105007)

Needs of AMI :
* Optimizes assets and operates efficiently.
* Accommodates all generation and storage options
* Provides power quality for 21st century needs.

Benefits of AMI :
* Improving the accuracy and flexibility of billing.
* Accommodating faster service restoration.

Standards of AMI :
* Multimedia and Data transmission are the standards of AMI.
IEC 61850 : Substation automation, integration of RES and SCADA.



- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ✓ The students are happy about the activity since they are able to give an immediate feedback about the concepts they learnt in the classroom.

  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ The students are made to comprehend the learned concepts inside the classroom itself after the classes over. It made them to remember the points very well.
    - ✓ The practice created a safe and open space for students to share their feedback. This can increase trust between me and my students because now they know that I am caring about their conceptual understanding of the subject taught.

  - ❖ ***Challenges faced in implementation:***
    - ✓ Few students are not interested to participate in the activity due to their lacking ability to comprehend and present their concepts learnt.
    - ✓ The time management for the activity is an issue. I could not able to complete the activity by planned time.

References:

- ❖ I.S. Jha, Subir Sen, Rajesh Kumar, and D.P. Kothari, "Smart Grid Fundamentals & Applications", New Age International Publishers, 1st Edition, 2019.
- ❖ Bharat Modi, Anu Prakash, and Yogesh Kumar, "Fundamentals of Smart Grid Technology", S. K. Kataria & Sons, 1st Edition, 2019.
- ❖ https://nptel.ac.in/courses/108107113 (Lecture Nos. 5 & 6).

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Innovative Practice Description

- **Unit / Topic: Unit IV / Power Quality issues of Grid Connected Renewable Energy Sources**
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 16
- **Activity Chosen: Ungraded Quiz**
- **Justification:**
  - The chosen topic deals with power quality issues present in the grid and its mitigation methods. Usually the students will find it difficult to understand the methods and also to properly address the power quality issues arises in the grid while interconnecting renewable energy sources to it. By conducting this ungraded quiz, I helped them to understand the concepts better.
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
  - The students are taken to the Power system simulation laboratory and each one is occupying the one system with Internet connectivity.
  - The quiz has been created in the Quizizz online tool and pass code has been shared with the students.
  - All the students have started the quiz at the same time. At the end of the quiz, the correct answers will be shown to them in the screen.
  - Finally, I explained the correct answer for the quiz questions with proper reasons.
- **CO – PO / PSO mapping:**

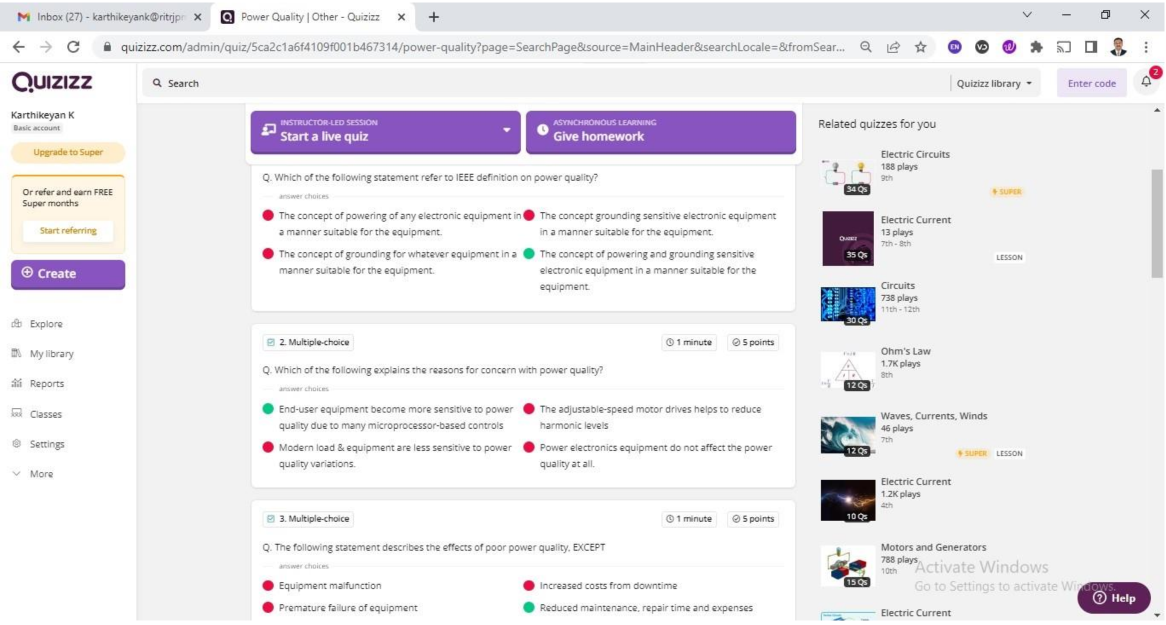
CO	PO 1	PO 2	PO 5	PO 8	PO 12	PSO 1
CO2	3	1	1	1	1	1

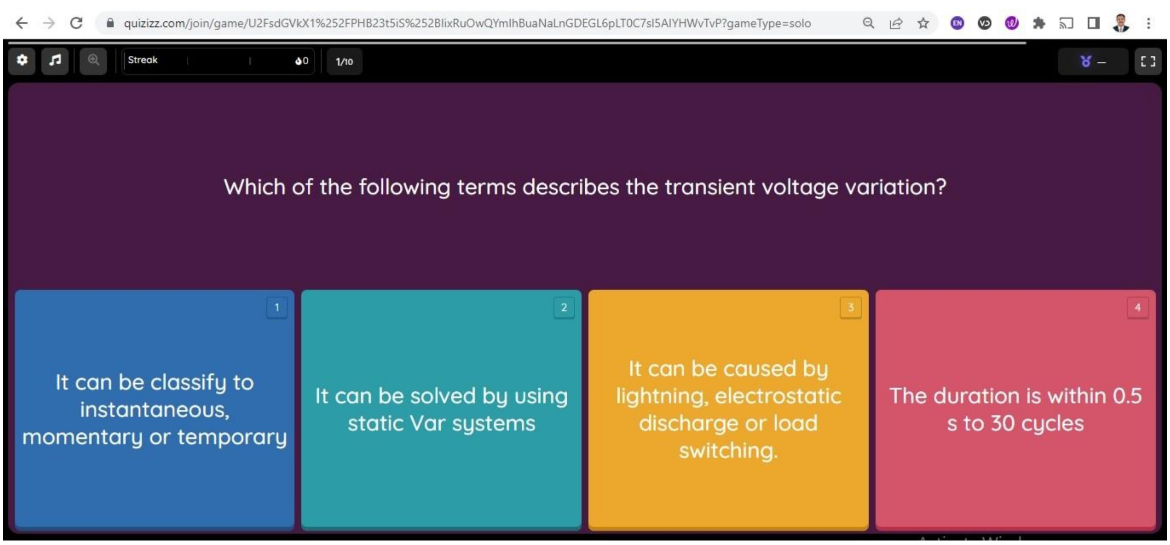
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 5**	**PO 8**
	1	**1**
Justification for correlation	The students are using online tool for doing this quiz. Hence it is mapped with this PO at level 1	The students are doing this quiz without discussing and copying from others. Hence it is mapped with this PO at level 1

- **Images / Screenshot of the practice:**





- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ✓ The students have enjoyed the quiz. They told they have been given with the chance of remembering the concepts in the same class itself.
  - ❖ ***Benefit of the practice:***
    - ✓ The students are getting the feel of involved in the learning process.
    - ✓ Since the quiz is not included in the marks calculation, the students have done the quiz without any test fear. Hence, they participated freely.
  - ❖ ***Challenges faced in implementation:***
    - ✓ Few students have not much interest in attending the quiz sincerely, since the quiz is ungraded. Managing those students and making them to do the quiz is difficult task for me in this activity.

References:

- ❖ I.S. Jha, Subir Sen, Rajesh Kumar, and D.P. Kothari, "Smart Grid Fundamentals & Applications", New Age International Publishers, 1st Edition, 2019.
- ❖ Bharat Modi, Anu Prakash, and Yogesh Kumar, "Fundamentals of Smart Grid Technology", S. K. Kataria & Sons, 1st Edition, 2019.
- ❖ https://article.sciencepublishinggroup.com/html/10.11648.j.ijepe.s.2015040501.14.html
- ❖ https://www.icrepq.com/icrepq'10/505-Khadem.pdf

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Innovative Practice Description

- **Unit / Topic: Unit V /** Communication channels
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 20
- **Activity Chosen:** Mind map
- **Justification:**
  - Mind mapping is a method for students to see the meanings of learning materials by portraying the connections between ideas, concepts and terms.
  - Mind maps are a cross-disciplinary, active learning technique that helps students manage concepts into sub-concepts, synthesize information, see a larger picture and develop higher-order thinking skills and strategies.
  - For faculty, mind maps provide a method to quickly understand students' thought processes and understanding of concepts.
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
  - The students were instructed to draw the mind map after lecture was over using any free mind mapping software available in the internet.
  - The selected few students were asked to show their mind map in the LCD projector and explain their map and their understanding about the concept.
  - The instructor helped the students for better understanding by adding the points.
- **CO – PO / PSO mapping:**

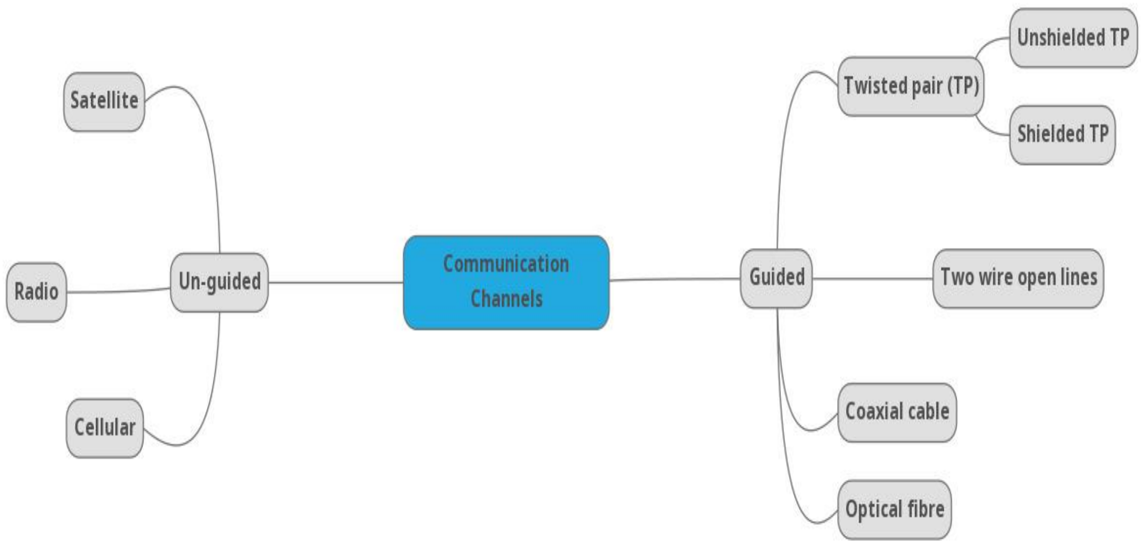
CO	PO 1	PO 2	PO 5	PO 8	PO 12	PSO 1
CO5	3	1	1	1	1	1

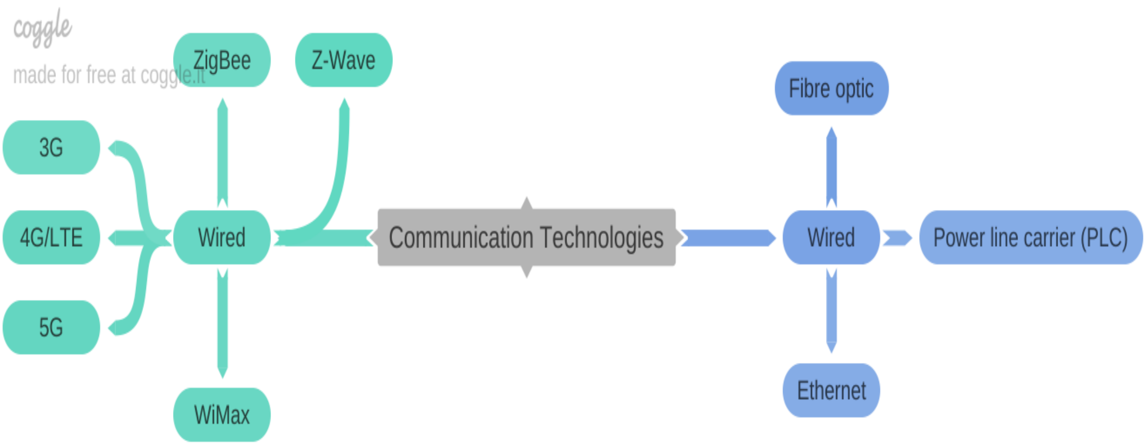
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 1	PO 5	PO 10
	1	**1**	**1**
Justification for correlation	The students know basics of the communication medium and its mode of operation for drawing the map. Hence they are able to visualize complex concepts in simple maps.	The students will use latest drawing software to draw the map. Hence they are using modern tools for learning the concepts.	The students are able to express the learning through the mind map. Hence the communication is getting improved.

- **Images / Screenshot of the practice:**





- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - ✓ The student shown much interest in using the special software available freely on the internet. The started using their chosen software for creating mind map not only for this topic but also for other similar topics.

  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ The pictorial representation of the complex concepts has helped the students to remember and recall the concepts easily.
    - ✓ The activity has helped them to apply the technique for other new topics. It helped them to learn new concepts.
    - ✓ The students expressed that the activity made them to understand the complex ideas easier.
    - ✓ It improved the creativity and presentation skills of the students.

  - ***Challenges faced in implementation:***
    - ✓ Few students are not able to identify the software and how to use them for creating mind map. I assisted them in doing so. This caused me to take more time in completing the exercise.
    - ✓ Few students are not able to visualize the concepts as a mind map. I helped them to think and draw by making complex concepts simple sub concepts.

References:

- I.S. Jha, Subir Sen, Rajesh Kumar, and D.P. Kothari, "Smart Grid Fundamentals & Applications", New Age International Publishers, 1st Edition, 2019.
- Bharat Modi, Anu Prakash, and Yogesh Kumar, "Fundamentals of Smart Grid Technology", S. K. Kataria & Sons, 1st Edition, 2019.
- https://www.mindmeister.com/blog/mind-mapping-benefits-who-needs-mind-maps/ https://nulab.com/learn/strategy-and-planning/8-science-backed-benefits-of-mind-mapping/

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